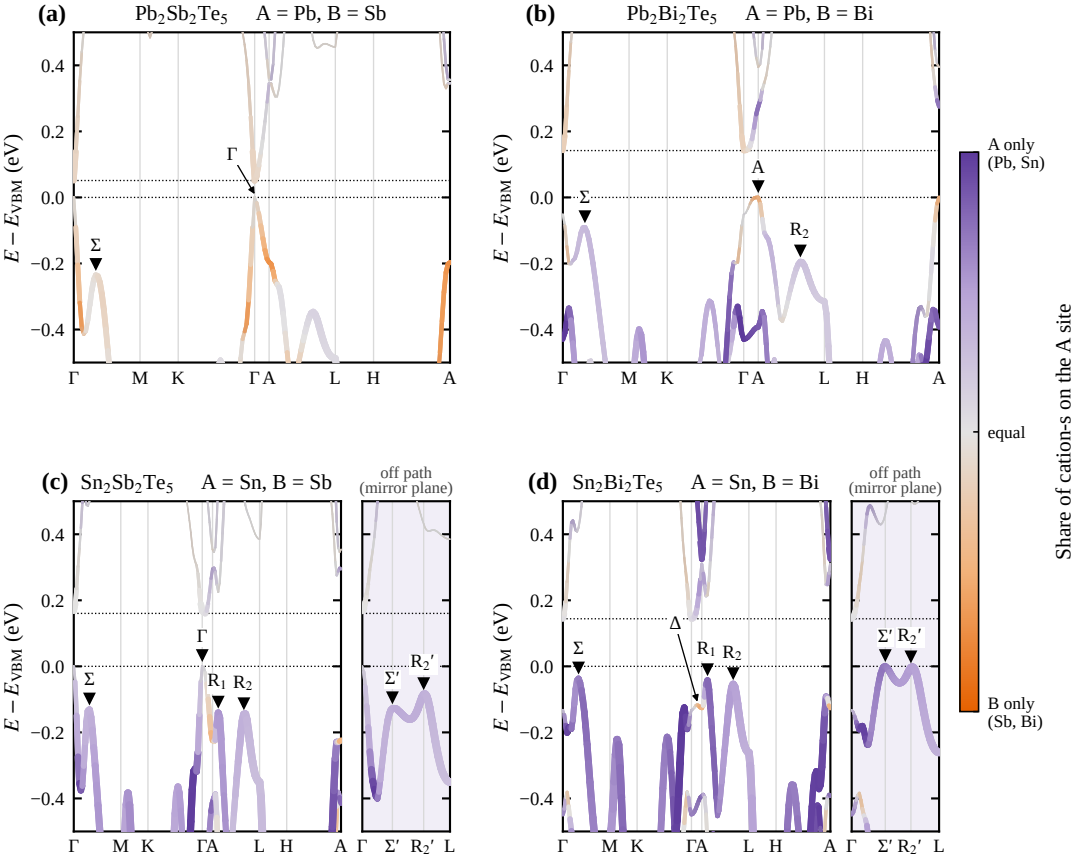

(a)
$Pb_2Sb_2Te_5$ A = Pb, B = Sb
(b)
$Pb_2Bi_2Te_5$ A = Pb, B = Bi
(c)
$Sn_2Sb_2Te_5$ A = Sn, B = Sb
(d)
$Sn_2Bi_2Te_5$ A = Sn, B = Bi
$E - E_{VBM}$ (eV)
0.4
0.2
0.0
−0.2
−0.4
Γ
M
K
Γ A
L
H
A
Σ
Γ
A
Σ
R2
Σ
Γ
R1
R2
Σ′
R2′
Σ
Δ
R1
R2
Σ′
R2′
off path
(mirror plane)
Σ′
R2′
Γ
Σ′
R2′
L
A only
(Pb, Sn)
equal
B only
(Sb, Bi)
Share of cation-s on the A site